

Math 514, F19
Quiz 8

Name: Answer

Instructions: You must show all of your work to receive credits for correct answers.

Problem 1. (4 points) The current price of a security is 40. Given a nominal interest rate of 6%, compounded continuously, find a lower bound for the price of a call option that expires in six months with a strike price of 38.

$$S + P - C = Ke^{-rt} \leftarrow \text{The put-call option parity formula.}$$

$$\text{Here } r = 0.06, S = 40, K = 38$$

$$C = S + P - Ke^{-rt}$$

Since $P \geq 0$, we have

$$C \geq S - Ke^{-rt} = 40 - 38e^{-0.06 \times (\frac{1}{2})} \rightarrow 6 \text{ months} = \frac{1}{2} \text{ year}$$

$$\approx 3.123$$

Problem 2. (6 points) An European call and put option on the same security both have the expiration time $t = 2$ years and the strike price $K = 25$. The call option sells for 3 (i.e. $C(25, 2) = 3$) and the put option for 6 (i.e. $P(25, 2) = 6$). Assume the nominal continuously compounded interest rate is 10% and the current stock is 20. Identify an arbitrage.

$$S + P - C = 20 + 6 - 3 = 23$$

$$Ke^{-rt} = 25 \times e^{-0.1 \times 2} \approx 20.468$$

$$\text{Thus } S + P - C > Ke^{-rt}$$

An arbitrage can be obtained by selling the security short, selling a put option and buying a call option:

$$\text{Initial yield: } 20 + 6 - 3 = 23$$

$$\text{Cost at } t = 2 \text{ years: } 25$$

$$\begin{aligned} \text{The present value of the gain is } & 23 - 25e^{-0.1 \times 2} \\ & = 23 - 20.468 \\ & = 2.532 \end{aligned}$$